Sara Logsdon

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Education

University of Pennsylvania — Philadelphia, PA
Doctor of Philosophy (PhD) Mathematics — Expected 2030

University of Georgia, Morehead Honors College — Athens, GA
Combined Master of Arts (MA) and Bachelor of Science (BS) in Mathematics, Summa Cum Laude — 2025
Master's Thesis Advisor: Pete L. Clark
Master's Thesis Title: Polynomial Functions over Finite Rings

Budapest Semesters in Mathematics — Budapest, HU
Mathematics courses with honors — January 2024–May 2024

University of Oxford, Maymester, Keble College — Oxford, UK
Philosophy of Language — May 2022

Work Experience

NASA Goddard Space Flight Center — Greenbelt, MD
Space Systems Quality Engineering Intern — May 2026-August 2026

- Developed formal verification demonstrations using α, β-CROWN and Marabou to verify safety and robustness properties of neural networks relevant to NASA applications
- Built theorem-proving demonstrations in PVS and NASALib using Plaidypvs, exploring their use in the verification of AI-enabled systems

Johns Hopkins Applied Physics Laboratory — Laurel, MD
Combat Systems and Analysis Intern — May 2023–August 2023, May 2024–August 2024

- Implemented network topology algorithms in MATLAB and C++ to a United States Navy project, enabling efficient communications between ships and planes
- Wrote original single and multi-beam scheduling algorithms for the U.S. Navy using linear optimization in Python

Department of Energy Idaho National Laboratory — Idaho Falls, ID
Cybersecurity Intern — June 2022–December 2023

- Analyzed formal methods applied to cyber-physical models of critical infrastructure and the cyber-physical critical security and resilience properties of such models and infrastructure
- Drafted white paper for formal methods-based approach to security using homomorphic encryption and theorem provers
- Explored the use of SAT/SMT solvers for toy problems such as Rubiks cubes and sudoku puzzles

Star Tutors — Remote
Mathematics Tutor — September 2019–May 2025

- Tutored high school students in algebra, calculus and statistics

Honors and Awards

National Defense Science and Engineering Graduate (NDSEG) Fellowship (2025-2028)

National Science Foundation (NSF) Graduate Research Fellowship Program, Honorable Mention (2025)

Joy P. Williams Physical Sciences Award (2025)

- Selected by the Morehead Honors College; awarded to an outstanding female student studying physical sciences

Barry M. Goldwater Scholarship (2024)

Charles M. Strahan Award (2024)

- Selected by University of Georgia mathematics faculty; first place among all third-year mathematics majors

University of Georgia Libraries Undergraduate Research Award, 1st Prize (2024)

- Awarded to my paper "On the Group Structure of Conic Sections and Elliptic Curves" for excellence in student inquiry

University of Georgia Center for Undergraduate Research Award (Summer 2022, Fall 2022, Spring 2023)

- Awarded by University of Georgia's Center for Undergraduate Research Organization to top undergraduate researchers

Hollingsworth Award (Fall 2021, Spring 2022)

- Top student in select early upper division math courses (awarded for MATH 3500 and MATH 3510)

University of Georgia Foundation Fellowship (2021–2025)

- University of Georgia's premier undergraduate merit-based scholarship, covering full cost of attendance, research and conference grants, and spring travel-study

Publications and Presentations

Publications

- S. Logsdon, *Unlocking Sudoku's Secrets.* Chalkdust Magazine, Issue 21. https://chalkdustmagazine.com/features/unlocking-sudokus-secrets (2025).
- S. Logsdon, *Exploring Applied Cryptosystems to Formally Verify Security in Cyber-Physical Systems.* U.S. Department of Energy Office of Scientific and Technical Information. https://www.osti.gov/biblio/1880065 (2022).

Submitted

- S. Logsdon, *On the Group Structure of Conic Sections and Elliptic Curves.* Under review for publication at: Rose-Hulman Undergraduate Mathematics Journal.

Preprints

- S. Logsdon, I. Miklós, A. Maheshwari, A. Zhang, *A Dichotomy Theorem on the Complexity of 3-Uniform Hypergraphic Degree Sequence Graphicality.* https://arxiv.org/abs/2411.19049. (2024)
- S. Logsdon, *Fully Homomorphic Encryption: A Mathematical Introduction.* https://eprint.iacr.org/2023/1402 (2023).

Presentations

- S. Logsdon, A. Zhang, *A Dichotomy Theorem on the Complexity of 3-Uniform Hypergraphic Degree Sequence Graphicality.* Presented at: 2025 Joint Mathematics Meeting, Seattle, WA, January 2025.
- S. Logsdon, *Deformation Quantization for Creating Non-Commutative Algebras.* Presented at: University of Georgia Algebraic Geometry Seminar, Athens, GA, October 2024.
- S. Logsdon, D. Ott, V. Schneider, *Stochastic Unitary Constraints.* Presented at: SciPy Scientific Computing with Python, Austin, TX, July 2023.
- S. Logsdon, *Group Theory and Geometry.* Presented at: University of Georgia Mathematics Directed Reading Symposium, Athens, GA, December 2022.

Skills

Java, Python, MATLAB, Wolfram Mathematica, Git, Linux, LaTeX